Follow the chain from force to position

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Force and motion / ES: fuerza y movimiento / PT: força e movimento

Net force → a → velocity and position.

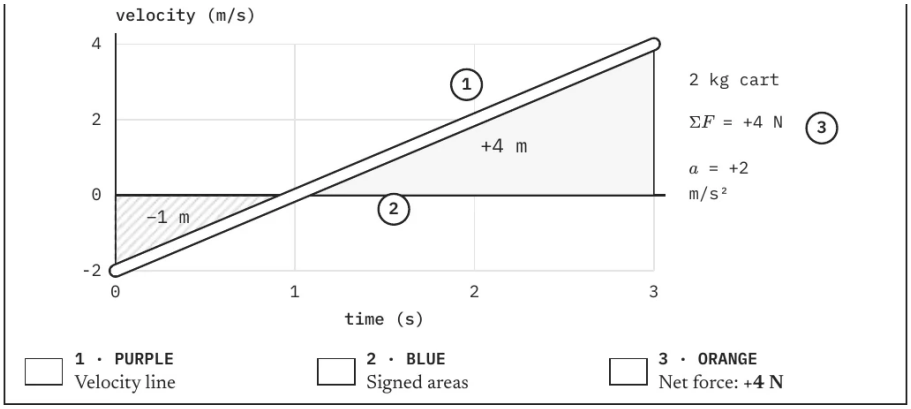
Also on your note page: Interval, Reversal, Area.

B READ THE RELATIONSHIP

Force → acceleration → velocity → position

As you read, find: force · velocity · position.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 2 kg cart starts at 1 m with velocity −2 m/s . A constant +4 N net force acts for 3 s . Find final velocity and position.

Given m = 2 kg ; Σ F = +4 N x 0 = 1 m ; v i = −2 m/s t = 3 s

Rule Find a from force. Then steady-a equations.

Work a = 4/2 = +2 m/s² v f = −2 + 2(3) x f = 1 − 6 + 9

Answer +4 m/s ; +4 m Δ x = +3 m ; distance = 5 m .

E SAVE FOR THE FINAL MEETING

XJ's forecast needs its present motion and the forces changing it. Use one reference frame through the entire calculation.

Read the physics: Force and motion

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Divide net force by mass to find acceleration. For constant acceleration, use the initial velocity and elapsed time to find final velocity.

02 HOW IT WORKS

Use the signed area under the velocity–time graph for displacement. Add displacement to starting position. A section below zero subtracts from displacement but still counts toward distance.

03 WHAT TO CHECK

With zero net force a moving object continues at its initial velocity. With constant nonzero net force its velocity changes by the same amount each second.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 2 kg cart starts at 1 m with velocity -2 m/s . A constant $+4$ N net force acts for 3 s . Find final velocity and position. $+4$ m/s ; $+4$ m $\Delta x = +3$ m ; distance $= 5$ m .

Model check: Force → acceleration → velocity → position.

05 CONNECT TO XJ

XJ's forecast needs its present motion and the forces changing it. Use one reference frame through the entire calculation.

SAY IT BEFORE YOU WRITE

What do you need besides force and mass to predict position? From force I can find ___. To find position I also need ___.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Force and motion / ES: fuerza y movimiento / PT: força e movimento

Say it in any language. Draw or use a few words. Use the notes to check your idea.

Meaning	How can you tell?
To me, force and motion means …	A cart feels a net force to the right. What else do you need to tell whether it is speeding up? I also need the direction of the cart's ___.
Example or sketch	**Non-example**
I see it when …	This looks close, but …

B DRAW WITH YOUR TEACHER

Build a four-frame chain from force to position. Add the missing inputs and units beside the arrows.

Labels you can use: force · velocity · position.

force | acceleration | velocity | position

Tell a partner: From force I can find ___. To find position I also need ___.

On assignment: Force and motion

SECTION E / BACK / FINAL MEETING ONLY Name _____

A telescope manager wants a final position. The telemetry feed contains net force and mass, but no starting velocity, starting position or elapsed time.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Mission systems engineer**

Mark which part of the forecast you can release now. Send the operator a request for the missing measurements.

☐ **News assignment editor**

Prepare a "known / still needed" graphic. Tell the anchor which result can be reported now and which must wait.

Word bank: force · velocity · position.

Try: "Our desk recommends ___. This mark shows ___."

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

"We first said ___. Now we ___ because ___."

Editor's check: Explain why a reversal matters when reporting total distance.